

Multiscale polymorphic uncertainty quantification based on physics-augmented neural networks

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ARTICLE INFO

Keywords:

Homogenization
Uncertainty quantification
Multiscale analysis
Multiscale uncertainty quantification
Polymorphic uncertainty
Physics-augmented neural networks.

ABSTRACT

With this contribution, the aim is to incorporate and evaluate the uncertainty in multiscale structural analyses. The material properties of composites (e.g., concrete, spinoidal structures) consequently depend on structural parameters and actual realizations of the composite mesostructures. Uncertainties on the mesoscale lead to uncertain behavior on the macroscale. Based on scale separation and following the current homogenization methods, a surrogate model is introduced, which enables the uncertainty quantification of macroscopic structures based on uncertainties at the mesoscale. Through the usage of Neural Networks (NN)s as surrogate models for the composite material, sampling-based uncertainty quantification schemes are enabled in large elastic deformations. A formulation of NNs that incorporates physical information of hyperelastic materials in the network structure is used and expanded with uncertain parameters to further reduce the information needed for the training of the NN. The proposed procedure enables the consideration of aleatoric, epistemic, and polymorphic uncertainty. For the training of the NN, a domain separation is proposed, which allows the efficient pre-training of the neural network.

1. Introduction

In computational mechanics, structural behavior is represented by different numerical models. To assess structural behavior, assumptions about material models are required to represent the stress-strain relationship of the investigated structure. If composite materials are considered, such materials consist of several components that are combined in a specific way to create a structure combining the microscopic material properties of the components, so that the combined structure displays specific, designed properties. Thus, the component material properties, as well as the arrangement of those components, influence the resulting material properties. Usually, the microscopic material properties of the components are known, while the influence of the structure is unknown. The resulting properties through the combination of the components are thus dependent on the arrangement of its components and evaluated by solving a boundary value problem over a representative part of the composite material. This representative part is called a *representative volume element* (RVE). The most frequently used finite element scheme used to solve the macroscopic structure is called FE^2 , which is numerically costly. In each of the components of the RVE and even the structure itself, naturally, uncertainties are contained [1–3]. Suppose uncertainty quantification is conducted to assess the influence of the uncertainty displayed by an RVE on the macroscopic results. In that case, the numerically costly solution of repeated FE^2 evaluations leads to high numerical costs in traditional uncertainty quantification schemes [1]. Thus, different strategies are developed to reduce the numerical effort of

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uncertainty quantifications with multiscale tasks.

In several contributions, machine learning methods are utilized to learn the stochastic response of the composite material, based on the composite's positions in the macroscopic FE. This is realized in [1,4] through GAUSSIAN process regression and in [5] by polynomial chaos expansion. In [6], the stress-strain relation of linear elastic small deformation-problems is approximated by *recurrent neural networks*, while in [7] neural networks are utilized to directly approximate the macroscopic energy of uncertain RVEs. Also, reduced order models are utilized to approximate the RVE responses, see [8]. Alternatively, several methods are documented that consider fuzzy, stochastic, or polymorphic uncertainty through adaptation of the FE² solution schemes, to increase the efficiency of the quantification schemes [9–11]. In other advancements, the macroscopic boundary value problem is not solved by an FE²-scheme or its approximation, but by evaluation of the boundary value problem with a *data driven* solution scheme. The data for this data driven solution scheme is obtained by evaluating the uncertain stress-strain relation of the RVE as in [12].

In the presented literature, methods for uncertainty propagation within a multiscale analysis are introduced. They are either applicable to small deformations [6–8,11], their solution is specific for a macroscopic task [1,4,5] or a specific type of uncertainty [9,11]. Thus, within this contribution, a framework is introduced, which enables the efficient evaluation of uncertainties in large deformations with hyperelastic materials, applicable to different solution schemes for different types of uncertainty. To do so, neural networks are implemented, which approximate the stress-strain relation of the composite, based on uncertain parameters. In order to reduce the amount of training data and enable tasks in large deformation frameworks, the formulation of *physics-augmented neural networks* (PANN)s [13] is investigated, which considers physical principles in the design of the NN. To enable uncertainty quantification, the PANN is extended by the uncertain parameters. A data generation scheme is presented, which allows differentiation between the domain of uncertain parameters and the mechanical domain, so training data are sampled efficiently. The contents of this contribution are structured as follows. First, in Section 2, the theoretical basis in continuum mechanics for the contribution developments is introduced. Following that, in Section 3, the theoretical basis in polymorphic uncertainty of parameters is outlined. In Section 4, the method of numerical homogenization under data uncertainty is introduced. Therefore, the metamodeling approach of PANNs is presented in an extended form incorporating polymorphic uncertainty. Additionally, supplementary methods are described for the sampling scheme, the space separation (uncertain parameter space and mechanical space), and the concept of confidence intervals as result quantities, which are extracted from p-boxes as polymorphic uncertain result quantities. In Section 5, two engineering structures are evaluated as numerical examples, and the results are discussed in order to demonstrate the applicability of the proposed methods. Concluding remarks are drawn in Section 6.

2. Theoretical basis of continuum solid mechanics

2.1. Kinematics

In the following contribution, a material body B and its domain in the current configuration $B \subset \mathbb{R}^3$ in the EUCLIDEAN domain are considered. Its motion from the bodies state at time $t = 0$ in the reference configuration $B_0 \subset \mathbb{R}^3$, to its current configuration B_t , at time $t \in \mathcal{T} := \{\tau \in \mathbb{R}_+ \mid \tau \geq t_0\}$, is described by a bijective mapping φ , that maps reference coordinates of a material point $\mathbf{X} \in B_0$ from the reference configuration onto their coordinates in the current configuration $\mathbf{x} \in B_t$. This mapping is defined as

$$\varphi : B_0 \times \mathcal{T} \rightarrow B_t : (\mathbf{X}, t) \mapsto \varphi(\mathbf{X}, t) = \mathbf{x}, \quad (1)$$

compare [6,14,15]. The corresponding displacement vector is introduced as $\mathbf{u} := \mathbf{x} - \mathbf{X}$. Thus, the deformation gradient $\mathbf{F}$ and the JACOBI determinant $J \in \mathbb{R}_+$ are defined as

$$\mathbf{F}(\mathbf{X}, t) = \frac{\partial \varphi(\mathbf{X}, t)}{\partial \mathbf{X}}, \quad J = \det(\mathbf{F}) > 0, \quad (2)$$

in accordance with [15]. A deformation measure free of rigid body motions is given by the right CAUCHY-GREEN deformation tensor $\mathbf{C} = \mathbf{F}^T \cdot \mathbf{F}$.

2.2. Stress measures

The CAUCHY-stress tensor, which is also referred to as tensor of true stresses, is denoted as $\boldsymbol{\sigma}$. The CAUCHY stress theorem postulates the stress vector $\mathbf{t} = \boldsymbol{\sigma} \mathbf{n}$ at the surface of the body ∂B_t in the current configuration by utilizing the current outward normal $\mathbf{n}$. Thus, an infinitesimal surface force $\mathbf{t} da$ is given, where da denotes an infinitesimal surface area. The 1st and the symmetric 2nd PIOLA-KIRCHHOFF stress tensors $\mathbf{P}$ and $\mathbf{S}$ are defined in [15,16] as

$$\mathbf{P} = J \mathbf{F}^{-1} \cdot \boldsymbol{\sigma} \quad \text{and} \quad \mathbf{S} = \mathbf{F}^{-1} J \cdot \boldsymbol{\sigma} \cdot \mathbf{F}^{-T}. \quad (3)$$

Thus, $\mathbf{P}$ is defined with respect to current and reference configuration, whereas $\mathbf{S}$ is completely defined with respect to the reference configuration.

2.3. Hyperelasticity

Within this contribution, the constitutive behavior of materials is restricted to *hyperelasticity*. The description for the stress-strain relation at a material point in hyperelasticity is defined indirectly by a hyperelastic potential, the strain energy density function

Ψ . To avoid stress-inducing rigid body motions, Ψ is assumed to depend on $\mathbf{C}$, which in turn is a function of $\mathbf{F}$. Hence, the statement $\Psi = \Psi(\mathbf{C}(\mathbf{F}))$, $\Psi \in \mathbb{R}_+$ follows as described in [15]. By evaluating the CLAUSIUS-PLANCK inequality, the relations for the stress measurements as introduced in Section 2.2 with respect to the hyperelastic potential follow, according to [13,15,17], as

$$\mathbf{P} = \frac{\partial \Psi}{\partial \mathbf{F}} \quad \text{and} \quad \mathbf{S} = \mathbf{F}^{-1} \cdot \frac{\partial \Psi}{\partial \mathbf{F}}. \quad (4)$$

In the following, the strain energy density is formulated with respect to the invariants of the right CAUCHY-GREEN deformation tensor $\mathbf{C}$ and some structural tensors $\mathbf{M}$, where the structural tensors describe assumptions about material symmetry. Therefore, the free strain energy density is expressed as

$$\Psi : \mathbb{R}^m \rightarrow \mathbb{R}, \quad \mathbf{I} \rightarrow \Psi(\mathbf{I}), \quad (5)$$

in accordance with [13,16]. The set of invariants $\mathbf{I} := (I_1, I_2, \dots, I_m) \in \mathbb{R}^m$ denotes an m -tuple containing a set of irreducible invariants $I_\alpha(\mathbf{C}, \mathbf{M})$. The size of the tuple is based on the assumption of the material symmetry and the corresponding invariants needed to describe it. By doing so, Eq. (5) fulfills several properties of hyperelasticity. Those are stated in [13,17]. Following from this formulation, the second PIOLA-KIRCHHOF tensor is derived from the free strain energy density with respect to the invariants of $\mathbf{C}$ by

$$\mathbf{S} = 2 \frac{\partial \Psi}{\partial \mathbf{C}} = 2 \sum_{\alpha=1}^m \frac{\partial \Psi}{\partial I_\alpha} \frac{\partial I_\alpha}{\partial \mathbf{C}}. \quad (6)$$

2.4. Numerical homogenization

If heterogeneous materials are considered, an appropriate approximation of their behavior is realized by splitting the boundary value problem into a problem on multiple scales. A homogeneous material description on the *macroscale* and inhomogeneous ones on lower scales are formulated. The composite material behavior is evaluated by solving the boundary value problem on the mesoscale, which entails the characteristic properties of the composite. All properties of the composite's components are assumed to be known, see [18]. To assess the interaction of the composite's components, they are homogenized. With that, the characteristic behavior of the composite material is obtained [19].

The macroscale is described with some body $B_0^{\text{macro}} \subset \mathbb{R}^3$ with a characteristic length $l_{\text{macro}} \in \mathbb{R}_+$. Further on in this contribution, only a single lower scale called *mesoscale* is considered. In order to describe the heterogeneous material behavior on the mesoscale, a second boundary value problem with a characteristic length $l_{\text{RVE}} \in \mathbb{R}_+$ is formulated. This is done due to the fact that the inhomogeneous material is considered homogeneous if the heterogeneities are far smaller than the evaluated structure itself. This is expressed through the characteristic lengths as

$$\delta = \frac{l_{\text{macro}}}{l_{\text{composite}}}, \quad \text{if } 1 \gg \delta > 0, \quad (7)$$

compare [20,21]. Thus, at each material point on the macroscale, the mesoscale continuum is evaluated in order to obtain the composite material behavior. As approximation of the mesoscale behavior, a *volume element* (VE) is constructed. Effective macroscopic properties are obtained by calculating the volume average of the respective quantity $\square$ over the VE-domain, as in [18,20], by

$$\bar{\square} = \frac{1}{V_{\text{VE}}} \int_{B_0^{\text{VE}}} \square dV_{\text{VE}}. \quad (8)$$

Thereby, V_{VE} is the volume of the VE and B_0^{VE} is its domain. The effective properties are dependent on the realization of the structure and, therefore, uncertain. In [19,22], the ergodicity of the structure is introduced as a property of the composite, stating that, based on $V_{\text{VE}} \rightarrow \infty$, the volume average of macroscopic quantities $\bar{\square}$ of a single realization of the mesostructure is representative for the ensemble of all possible mesostructures. Thus, if V_{VE} is large enough, a deterministic effective value, independent of the actual realization of the mesostructure, for the macroscopic quantity of interest is found, see [19]. A VE independent of the specific realization of its structure is called *representative volume element* (RVE), see [18,19]. Appropriate boundary conditions follow from the equivalence of the macroscopic energy at a material point and the volume average of the mesoscopic energy at the material points corresponding mesoscale. This is also known as HILL-MANDEL condition and defined in [20] as

$$\bar{\mathbf{P}} : \dot{\bar{\mathbf{F}}} = \overline{\mathbf{P} : \dot{\mathbf{F}}}. \quad (9)$$

From this condition, appropriate boundary conditions for the RVE evaluation are presented in [23]. Common ones are linear, traction, or periodic boundary conditions. Periodic boundary conditions state that opposing boundaries are connected with each other. They prescribe a periodic displacement $\mathbf{u} \in \mathbb{R}^3$, which is equal at opposing edges. From that, the traction $\mathbf{p} \in \mathbb{R}^3$ follows, which is antiperiodic on opposing edges. Hence, the periodic boundary conditions are defined, according to [6,14,23] as

$$\mathbf{u} = (\bar{\mathbf{F}} - \mathbf{1}) \cdot \mathbf{X}. \quad (10)$$

In [24,25], it is shown that even if the structure of the RVE is non-periodic, periodic boundary conditions still lead to convergence of effective macroscopic quantities. Due to the fact that convergence of effective macroscopic quantities is achieved with smaller volumes under periodic boundary conditions, they allow the evaluation of composites with less numerical effort than other boundary conditions [25]. Thus, lower numerical effort is required to compute the effective macroscopic properties of the RVE. Therefore, periodic boundary conditions are utilized in this contribution.

2.5. Physics-Augmented neural networks in homogenization schemes

Due to their nested solution scheme, multiscale simulations lead to high numerical effort. Thus, several methods exist to lower this effort by approximating parts of the solution scheme. For instance, in [26], the composite is simplified and surrogated by an analytical function.

Another method is surrogate models for the stress-strain relation, surrogating the RVE computations. By numerical material testing, the RVE is computed for the expected sampling domain (design of experiment), and the results are used to train a surrogate model as in [27]. By properly including physical information into the structure of a surrogate, the amount of required training data to describe the stress-strain relation is reduced significantly [17]. An approach is presented in [13,17], where artificial neural networks are used as such physics-augmented surrogates, to predict the strain energy function as relation between $\bar{\mathbf{F}}$ and $\bar{\mathbf{S}}$, which is defined as

$$\bar{\Psi} : \mathbb{R}^m \rightarrow \mathbb{R}_+, \quad \bar{\mathbf{I}}(\bar{\mathbf{C}}(\bar{\mathbf{F}}), \mathbf{M}) \mapsto \bar{\Psi}(\bar{\mathbf{I}}). \quad (11)$$

This is referred to as PANN, which is used to predict the indirect functional relation between $\bar{\mathbf{I}}$ and $\bar{\Psi}$. The training data set, used to train the PANN, is defined as $\mathcal{D} := (\mathcal{D}^1, \mathcal{D}^q, \dots, \mathcal{D}^Q)$ and consists of Q tuples described by the set $\{q \in \mathbb{N} \mid 1 \leq q \leq Q\} \subset \mathbb{N}$. Each tuple consists of the obtained invariants $\bar{\mathbf{I}}$, its corresponding stress state $\bar{\mathbf{S}}$ and is denoted as $\mathcal{D}^q = (\bar{\mathbf{I}}^q, \bar{\mathbf{S}}^q)$. Thereby, $\bar{\mathbf{I}}$ is obtained directly by evaluation of the relation $\bar{\mathbf{I}}(\bar{\mathbf{C}}(\bar{\mathbf{F}}))$. The training is conducted on the data set by evaluating the loss described in [17] by

$$\mathcal{L} = \frac{1}{Q} \sum_{q=1}^Q \|\bar{\mathbf{S}}_{\text{NN}}^q - \bar{\mathbf{S}}_{\text{RVE}}^q\|^2. \quad (12)$$

The neural network is trained indirectly by using Eq. (6) to reconstruct $\bar{\mathbf{S}}_{\text{NN}}^q$. The output of the PANN, the strain energy density, is indirectly included in the loss function through derivative calculation of the PANN according to Eqs. (6) and (12). For the evaluation of the loss, the gradients of the output are calculated. Therefore, the strain energy density does not need to be known. The obtained networks fulfill several physical conditions exactly by construction. These conditions are found in [13]. Thus, those networks are named *physics-augmented*. To incorporate the NN into an FE framework, the stiffness is evaluated as in [28] by

$$\begin{aligned} \bar{\mathbf{C}}^{\text{NN}} &= 4 \frac{\partial^2 \bar{\Psi}^{\text{NN}}}{\partial \bar{\mathbf{C}} \partial \bar{\mathbf{C}}} \\ &= 4 \sum_{\alpha=1}^m \sum_{\beta=1}^m \frac{\partial^2 \bar{\Psi}^{\text{NN}}}{\partial I_\alpha \partial I_\beta} \cdot \left(\frac{\partial I_\alpha}{\partial \bar{\mathbf{C}}} \otimes \frac{\partial I_\beta}{\partial \bar{\mathbf{C}}} \right) + 4 \sum_{\alpha=1}^m \frac{\partial \bar{\Psi}^{\text{NN}}}{\partial I_\alpha} \cdot \frac{\partial^2 I_\alpha}{\partial \bar{\mathbf{C}} \partial \bar{\mathbf{C}}}. \end{aligned} \quad (13)$$

Thus, the network itself needs to be twice differentiable and unbounded, and the activation functions used are required to obey these requirements.

3. Polymorphic uncertainty of parameters

In structural safety, uncertainties have to be considered, which significantly influence the results of a structural analysis. Those uncertainties are usually assessed through evaluation of a parametrized computational model. The model itself is defined as $\mathcal{M} : U \rightarrow Z$, $u \mapsto \mathcal{M}(u) = z$, where input parameters u lead to a response of $\mathcal{M}$, described by parameters z . If uncertain parameters are considered, u are samples from U^u , the space of the uncertain quantities evaluated. The inputs are propagated through $\mathcal{M}$, based on the type of uncertainty that is propagated. The resulting responses of $\mathcal{M}$ are then evaluated. In the following, basic uncertainty models and the concept of polymorphic uncertainty modeling are introduced, as well as random fields based on polymorphic uncertainty.

3.1. Randomness

For the description of aleatoric uncertainty, random quantities are furthermore denoted with a superscript $\square^r$. Based on a probability domain (Ω, Σ, P) , random characteristics are obtained. Ω is the sample domain. A random variable V^r is defined in accordance with [29] as the mapping

$$V^r : \Omega \rightarrow \mathbb{R} : \omega \mapsto V^r(\omega). \quad (14)$$

Furthermore, $P : \Sigma \rightarrow [0, 1]$ is the probability measure for the occurrence of an event $v \in \Sigma$, where Σ is the Σ -algebra. The corresponding *cumulative distribution function* (CDF) is subsequently defined as

$$F_{V^r}(v) = P(V^r \leq v) = F(v, \theta) \quad (15)$$

with a corresponding set of parameters θ . The distribution of random events for the specific task at hand is characterized by the CDF. In case of continuous random variables, the CDF is in relation to its *probability density function* (PDF) $f^r(v) = \frac{dF^r(v)}{dv}$.

If the parameters θ describing the CDF as in Eq. (15) are known, a set $\mathcal{A}_\beta = \{\omega \in \Omega \mid a_1 \leq V^r(\omega) \leq a_u\}$ containing a sample point obtained by the distribution function with a probability

$$P(\mathcal{A}_\beta) \geq 1 - \beta, \quad \beta \in [0, 1] \quad (16)$$

is called the confidence set of the corresponding distribution, as defined in [30]. The complementary probability β of a point depicts the probability of a random sample point being outside of $\mathcal{A}_\beta$. A quantile value q_p of a CDF is obtained by evaluating

$$q_p = F_{V^r}^{-1}(p), \quad p \in [0, 1]. \quad (17)$$

Within this contribution, confidence intervals are defined based on the assumption of symmetric probabilities of the peripheral bounds of the CDF by $\mathcal{A}_\beta = [q_{\frac{\beta}{2}}, q_{1-\frac{\beta}{2}}]$.

Uncertainty Quantification for Randomness. For the numerical simulations of random variables, uniform and independent random numbers are generated, which are then transformed to the predefined distribution as in [31], their inverse CDF is applied. Result quantities are then obtained by evaluating the response of $\mathcal{M}(U^r)$ about quantities of interest as in [32]. Approaches that follow this scheme are called MONTE-CARLO-based approaches.

3.2. Intervals

For the modeling of epistemic uncertainty, intervals are utilized. The occurrence of a certain event is assessed by a possibility measure. Interval-valued quantities are denoted by the superscript $\square^i$ and are defined by a lower and an upper boundary as

$$V^i = [\underline{v}, \tilde{v}] = \{v \in \mathbb{R} \mid \underline{v} \leq v \leq \tilde{v}\} \subset \mathbb{R} \quad (18)$$

according to [31,33]. Thus, each element v is rated binary by its possibility through a characteristic function

$$\xi_{V^i} : \mathbb{R} \rightarrow \{0, 1\}, \quad v \mapsto \begin{cases} 1, & v \in V^i \\ 0, & v \notin V^i \end{cases} \quad (19)$$

If multiple interval-valued input quantities are considered and no interactions are between them, a corresponding input domain U for the computational model is obtained by the CARTESIAN product of all the interval input quantities as

$$U = U_1^i \times \dots \times U_n^i \subset \mathbb{R}^n, \quad (20)$$

where the intervals are combined in a hypercube of dimension n .

Uncertainty Quantification for Intervals. As shown in [33], uncertainty quantification for a computational model with interval input quantities results in interval-valued outputs. Such interval analyses are performed by either interval arithmetic approaches or by solving the optimization problem of finding the minimum and maximum of the computational model $\mathcal{M}$ with respect to the possible values u of the interval input space U , see Eq. (20). Hence, this optimization leads to an interval-valued result quantity

$$Z^i = \left[\min_{u \in U} \mathcal{M}(u), \max_{u \in U} \mathcal{M}(u) \right]. \quad (21)$$

3.3. Parametric probability-boxes

In scenarios where data is sparsely available but knowledge of the distribution of that data exists, *parameterized p-boxes* V^{pb} are defined in order to model that uncertainty, as described in [34,35]. The distribution is predefined for the aleatoric uncertainty and additional epistemic uncertainty is modeled with interval-valued parameters $\theta_1^i, \dots, \theta_{n_\theta}^i$. Parameterized p-boxes can be defined by the set of all parametrized CDFs. This set is expressed with dependence on the bunch parameters θ by

$$F_{V^{\text{pb}}}(\nu, \Theta^i) = \{F_{V^r}(\nu, \theta) \mid \theta \in \Theta^i \subset \mathbb{R}^{n_\theta}\}, \quad (22)$$

where the interval input domain $\Theta^i = \theta_1^i \times \dots \times \theta_{n_\theta}^i$, see Eq. (20), as defined in [36]. A full description of the result quantity is obtained by finding the corresponding boundary distributions, which are defined in [34–36] as

$$\underline{F}(\nu) = \min_{\theta \in \Theta} F_{V^r}(\nu, \theta) \quad \text{and} \quad \tilde{F}(\nu) = \max_{\theta \in \Theta} F_{V^r}(\nu, \theta). \quad (23)$$

Uncertainty Quantification for Parametric P-Boxes. If p-boxes are considered as input quantities, their evaluation leads to interval-valued output quantities, based on information-reducing measurements applied on the parameterized CDF. The solution scheme for such a polymorphic uncertainty task is implemented by stacking the solution methods corresponding to the aleatoric and epistemic uncertain parts and solving them respectively with the fitting methods. In Fig. 1, a schematic, parameterized p-box is depicted. Thereby, the interval-valued solution $[F(v_k), \tilde{F}(v_k)]$ at a certain point v_k is depicted.

3.4. Interval probability-based random fields

Based on the common probability space (Ω, Σ, P) , a random function is defined in [37] as

$$H^{\text{rf}}(\gamma, \omega), \quad \text{with } \gamma \in \Gamma \subset \mathbb{R}^n \text{ and } \omega \in \Omega, \quad (24)$$

which results in a point specific random variable $H_{\gamma^\star}^{\text{r}}(\omega) = H^{\text{rf}}(\gamma^\star, \omega)$, at a fixed point $\gamma^\star$ of the observation points γ in the domain of observation Γ with n dimensions.

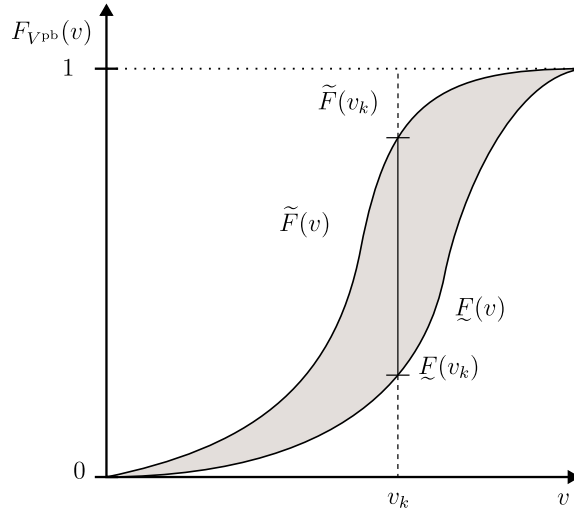


Fig. 1. Schematic depiction of a p-box with interval-valued probability $F_{Vpb}(v)$ at a particular point v_k .

In the following, only observation points in the EUCLIDEAN domain of the reference configuration (see Section 2) are considered, such that $H^{rf} = H^{rf}(\mathbf{X}, \omega)$. Therefore, the random process is called a random field. Moreover, the random field H^{rf} is called multivariate if it is a vector of random variables at each point in the observation space. The underlying CDFs are called marginal distributions, and different marginals

$$F_{H_i^{rf}} = P(H_i^{rf} \leq h_i) = F(h_i, \mathbf{X}, \theta_i) \quad (25)$$

are defined in accordance with [31,38]. If the random field is considered stationary, all marginals are independent of $\mathbf{X}$. Otherwise, a trend function of the random field is denoted as $E(H^{rf}(\mathbf{X}))$ and based on that function

$$\text{Cov}(\mathbf{X}_A, \mathbf{X}_B) = E(H^{rf}(\mathbf{X}_A), H^{rf}(\mathbf{X}_B)) \quad (26)$$

is defined as the autocovariance function between two points $\mathbf{X}_A, \mathbf{X}_B \in B_0$, as described in [38,39]. If one or more parameters θ of the random field are interval-valued, the corresponding marginal, as defined in Eq. (25), is a parametric p-box as defined in Eq. (22). Therefore, the resulting random field is an *interval probability-based random field* H^{iprf} , see [40]. Another possibility of defining such a random field is the interval-valued definition of the covariance function or its parameters in Eq. (26), as defined in [39]. The solution strategy is similar to as in Section 3.3. Samples of autocorrelated random fields are generated by a discretized KARHUNEN-LOÉVE-expansion, as described in [41].

4. Homogenization with structural uncertainty

The quantification of uncertainty is often conducted with sampling-based approaches [6]. Either optimization algorithms, used to compute response quantities of intervals, or MONTE-CARLO-based approaches, which are applied in order to evaluate aleatoric responses, are based on some sampling over the investigated computational model $\mathcal{M}$. The uncertainty is evaluated by sampling the uncertain parameters specifically for their algorithm. Then, the samples are propagated through the computational model to receive the uncertain result quantity or statistical estimators of result quantities. If uncertainty quantification is investigated for numerical homogenization of composites, $\mathcal{M}$ is a multiscale simulation. To evaluate the geometrical uncertainty of a mesostructure, the algorithm generating the mesostructure has to be parametrized, so sampling algorithms are capable of evaluating the uncertainty of the respective parameters.

4.1. Structural uncertainty between scales

If, at each GAUSS point of the macroscale boundary value problem, the stress-strain relationship obtained by evaluating the composite is dependent on uncertain parameters, the stiffness tensor is also dependent on uncertain parameters [5,7]. Thus, the response evaluated by the macroscale finite element scheme changes, based on its parametrization and therefore depends on the uncertain parameters. If parameters describing the structure of the RVE are modeled by uncertain variables, for each realization of uncertain parameters, a new RVE structure has to be generated. In this case, the convergence behavior of specific quantities of interest, e.g., $\bar{C}$, is altered by structural parameters of the RVE. This implies the evaluation of convergence for every realization of the RVE within the uncertainty quantification. Alternatively, prior to the FE simulation, an investigation of the influence of the uncertain parameters of the RVE on the convergence of effective macroscopic quantities against the volume average can be computed. Therefore, an RVE size large enough to calculate converged quantities independent of the domain of structural parameters is required.

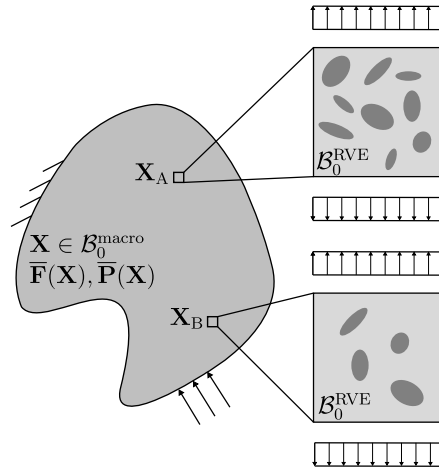


Fig. 2. Scheme of the multiscale boundary value problem – macroscale body B_0^{macro} (left) and uncertain structural inhomogeneities for arbitrary points X_A and X_B (right).

This ensures that the VE itself is representative of the quantity of interest for every possible realization of the RVE, based on the parametrization.

In this contribution, it is distinguished between *mesoscale uncertainty* and *macroscale uncertainty*. Assume that a VE, which is obtained by the structure generation algorithm, is an RVE. Then, it is independent of the exact realization of the structure as described in Section 2.4. Thus, for each RVE at each GAUSSIAN point of the macroscopic FE-simulation, the response of a specific RVE with fixed geometrical parameters is deterministic. Still, uncertainty in RVEs is possible through uncertain structural descriptors of the RVE and material descriptors of the base materials of the composite. This uncertainty is defined at the macroscale, even if the evaluation at the mesoscale and, therefore, the response of the RVE, is influenced by the uncertainty. This type of uncertainty is further called macroscale uncertainty, because it is defined on the macroscale boundary value problem. It is not reducible through enlargement of the VE. The space of a relevant support of all macroscopic input quantities, which parametrize the RVE and describe an uncertainty, is called $U \subset \mathbb{R}^u$. Thereby, the number of uncertain variables, which are considered, is described by u . Finding the relevant support is discussed in Section 4.2.

Also, this type of uncertainty may be spatially varying over the macroscopic structure. If those variations influence each other, their spatial variation is considered through correlation structures, e.g. modeled with random fields. Thus, the structural descriptors of a composite material vary over the macroscale, based on a correlation structure previously incorporated. In Fig. 2, a schematic description of the uncertain multiscale problem is shown. Uncertain boundary conditions on the macroscale, e.g., the exact shape of B , displacements, forces, etc., only influence $\bar{F}(X)$ and $\bar{P}(X)$.

Suppose that a VE is not representative. Then, for each fixed combination of geometrical parameters, its response is uncertain. This is called mesoscale uncertainty because the uncertainty stems from the mesoscale and is not describable by macroscopic input quantities. It is theoretically fully reducible by increasing the volume of the VE, so that an RVE is obtained. Suppose the material parameters of the composite materials are considered spatially dependent and uncertain. If such uncertainties are considered, they must follow the conditions stated in [19] to be considerable for mesoscale computations. Then, this spatial dependence on the mesoscale will be reduced to a fixed value described by macroscale parameters if only RVEs are considered. Thus, by homogenization, this type of uncertainty is considered an uncertainty in the structure of the RVE and, therefore, it will be reduced by choosing appropriately large VEs.

4.2. PANNs In multiscale uncertainty quantification

In Section 4.1, the specific requirements for the incorporation of uncertainty in RVEs, in order to evaluate uncertain structural parameters of those RVEs, are outlined. Due to the fact that multiple RVE evaluations are necessary within that framework and each FE^2 computation induces numerical effort, a quantification of uncertainty models described in Section 3 requires many numerically costly computations. Thus, sampling-based approaches that directly evaluate the parametrized RVE are numerically too expensive to compute [1]. Due to the fact that most of the numerical effort is induced by the generation, convergence test, and evaluation of the RVEs, the incorporation of surrogate models in place of the RVEs leads to acceptable numerical costs.

If random fields and other spatial uncertainties are considered in material models, their influence is considered by alterations to the sampling. At the GAUSSIAN points of the FE simulation, the samples are then used as input for the specific RVE. Thus, an efficient replacement of the parametrized RVE is achieved by using a parametrized surrogate for sampling-based uncertainty quantifications, which is faster than the RVE evaluation and still accurate. The utilization of neural networks in this context is advantageous. The advantage arises from the classical theory of multiscale analysis. Based upon [28], it is well established that the strain energy density, as well as its derivatives (stresses and material tangent), are an implicit function of the RVE geometry model. This fact is expressed

through Eq. (8), where the relation between macro- and microscale quantities is given by a volume integral. This relation is evaluated by the computation of result quantities based on a mesoscale boundary value problem, dependent on parameters of the RVE. Hence, the functional dependence is implicit. Nevertheless, a functional dependency is present, which is approximated advantageously by Neural Networks.

4.2.1. Uncertain quantities as additional input parameters of PANNs

Several methods for the prediction of the stiffness resulting from a certain strain state with a surrogate model exist [27,28,42–44]. In this contribution, the assumption of hyperelasticity is made. Based on this assumption, in [13,28] the concept of PANNs is laid out. Depending on the formulation of the NN and the mechanical assumptions, the input-output relation that needs to be learned changes. If the uncertainty quantification is based on an NN without any physical information, it requires many training data points and large NN structures in order to be approximated, due to the number of input dimensions and the possibly complicated relation based on the uncertain parameters and the mechanical input. The formulation for hyperelastic materials incorporating physical relations into the NN is used as described in [13,17,28] and expanded by incorporating the uncertain input variables U . This approach provides the advantage of physical information being included in the structure of the NN. By that, the information needed in order to describe the mechanical relation is reduced, which also leads to a reduced amount of overall information required in order to describe the stress-strain relation, based on uncertain parameters [17]. The energy at each point is dependent on the uncertain parameters, depicted by the RVE. The relation in Eq. (11) is expanded by the domain U , which is the space of descriptors the RVE is parametrized with, as introduced in Section 4.1. Thus,

$$\bar{\Psi} : \mathbb{R}^m \times \mathbb{R}^u \rightarrow \mathbb{R}_+ : (\bar{\mathbf{C}}(\bar{\mathbf{F}}), \mathbf{M}, \mathbf{u}) \mapsto \bar{\Psi}(\bar{\mathbf{I}}, \mathbf{u}) \text{ and } \mathbf{u} \in U \quad (27)$$

as a parameterized formulation of the strain energy function at the macroscale, is obtained. In this contribution, an approach is proposed to substitute the relation of Eq. (27) by a PANN, as an expansion to the method described in Section 2.5. With the presented expansion, the domain of the uncertain quantities U of the RVE are considered as additional input parameters of the neural network, which is reflected in the difference between Eqs. (11) and (27). Hence, the tuple of training data is expanded as $D^q = ((\mathbf{I}^q, \mathbf{u}^q), \mathbf{S}^q)$. If the second derivative of the PANN with respect to the invariants is dependent on $\mathbf{u}$, the relationship described by Eq. (27) can be incorporated into an FE framework as material law. The influence of parameters on the physical conditions of the PANN is discussed in [45,46].

Within a PANN, every physical information reduces the amount of training data [17]. Thus, the formulation with as much physical information as possible is advantageous. The first physical information is the formulation of the relation with respect to the macroscopic strain energy function $\bar{\Psi}(\bar{\mathbf{C}}, \mathbf{u})$. There, $\bar{\Psi}$ is dependent on $\bar{\mathbf{C}}$ as introduced in [47]. The procedure is advantageous if continuous changes in the approximated material classes occur based on the uncertain parameters. Arbitrary changes in material symmetries can be approximated with this formulation. A second physical information leads to the formulation as introduced in Section 2.5. It incorporates the information of the material class through the formulation of the energy density with respect to the corresponding invariants, see Eq. (27). Thus, the amount of information needed is further reduced. But the material symmetry needs to be investigated before training the PANN, and the PANN is incapable of predicting continuous changes based on the uncertain parameters. Because a formulation of the PANN as described with Eq. (27) reduces the amount of training data needed the most, it is the formulation pursued further. In Fig. 3, the changes in the uncertainty quantification by the proposed method are depicted. Therein, from the algorithm steps “training data generation” and “PANN training”, it is visible that many of the costly operations conducted within the original algorithm (Fig. 3, left), are computed before the uncertainty quantification part in the proposed method (Fig. 3, right).

Uncertain RVE material behavior is approximated by the parameterized PANN. The uncertainty of the mesoscale is approximated by the PANN through the domain of uncertain parameters U describing the corresponding uncertainty. Uncertainty quantification is conducted by evaluation of the uncertainty over the respective parameterized domain. In the case of aleatoric quantities, samples $\mathbf{u} \in U$ are drawn from the distributions of the respective quantity and evaluated by numerically calculating the response of the engineering model based on the samples as input quantities, as described in Section 3.1. In the case of epistemic quantities, as introduced in Section 3.2, the respective domain from U is evaluated through optimization schemes. Samples are drawn from U , the model's response to them is evaluated, and new samples are drawn afterwards. Polymorphic quantities combine those approaches. Thus, as long as the surrogate is trained in a reasonable domain, where samples are then actually drawn from uncertainty quantification based on that surrogate, it is enabled [6]. In contrast to [5,11], every type of uncertainty modeling on the macroscale (e.g., aleatoric, epistemic, polymorphic) is evaluable. Also, the PANN acts as a parameterized material model and is therefore independent of the position in $\mathbf{X}$. Thus, no limitations with respect to the kind of uncertainty (e.g., geometric imperfections, RVE parameter, forces, etc.) are present, overcoming limitations of the frameworks presented in [1,4,5]. The limitations of the proposed method are the incapability of approximating mesoscale uncertainty as introduced in Section 4.1 and the necessity for well-selected bounds of the sampling domain.

4.2.2. Training data acquisition

The input domain for the PANN consists of the domain $\mathcal{F} \subset \mathbb{R}^{3 \times 3}$, which describes the mechanical domain and contains all possible deformation gradients $\bar{\mathbf{F}}$ and the domain of uncertain parameters U , as introduced in Section 4.1. The resulting input domain Λ is a combination of $\mathcal{F}$ and U defined as

$$\Lambda = \mathcal{F} \times U \subset \mathbb{R}^{3 \times 3} \times \mathbb{R}^u. \quad (28)$$

In order to draw samples for the training of the PANN, samples are drawn from Λ , as defined in Eq. (28).

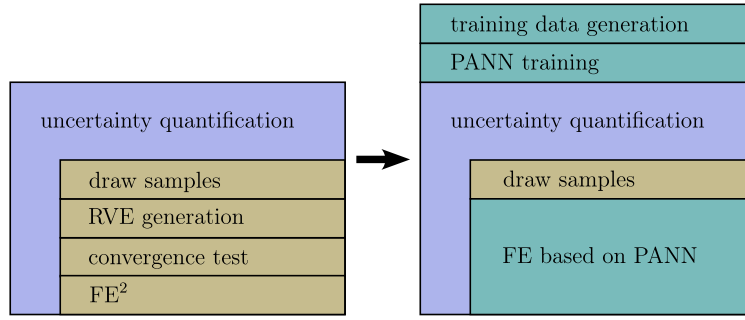


Fig. 3. Flow chart of the multiscale uncertainty quantification with FE^2 approach (left) and the proposed alternative flow chart with the PANN as surrogate for the mesoscale FE evaluation (right).

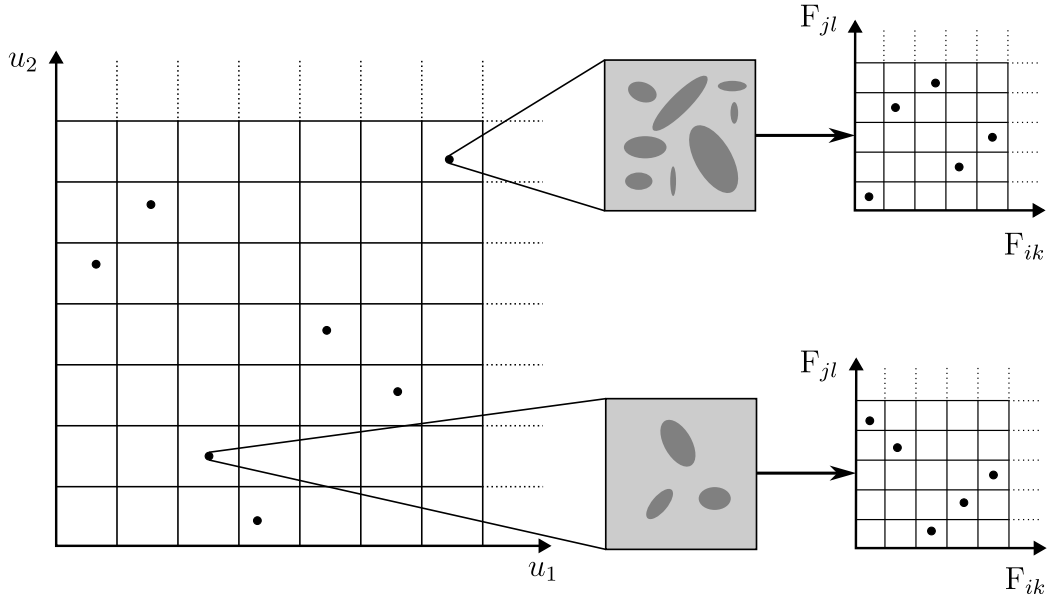


Fig. 4. Schematic description of the domain separation.

Space-filling designs, such as *latin hypercube sampling* (LHS) [48], draw samples in a manner such that the samples are space-filling and randomly distributed. Therefore, for each sample, a convergence test (see Section 4.1) has to be computed, which is numerically expensive for large sample sizes. In contrast to the mechanical parameters in $\mathcal{F}$, uncertain parameters, which describe geometrical properties of the RVE, require a new convergence test. Thus, the numerical effort from re-evaluating the convergence behavior of the quantity of interest stems from the space of uncertain parameters. To save numerical effort, a split of Λ into the domain of uncertain parameters $\mathcal{U}$ and the mechanical domain $\mathcal{F}$ is proposed. First, $\mathcal{U}$ is explored by a space-filling design. Second, for each of those samples, an RVE geometry is generated and tested for the convergence of the effective macroscopic properties (see Section 2.4). Third, for each generated RVE geometry, LHS samples are generated by space-filling design in $\mathcal{F}$. Fourth, for each of the samples in $\mathcal{F}$, the effective macroscopic properties of the RVE are computed.

By this procedure, a convergence test is required only for each sample in $\mathcal{U}$. Still, by the separation into distinct domains, the amount of required samples is overestimated and will not follow a space-filling design in Λ . This simplification is applied to reduce the numerical effort of the data generation. The space separation is depicted in Fig. 4. Therein, at two exemplary points from the LHS in $\mathcal{U}$, the corresponding parameters lead to different realizations of the RVE. For each structural realization (depicted by the two exemplary points), a separate LHS in the mechanical space is conducted. All entries of $\mathbf{I}$ and $\mathbf{U}$ are scaled in a range between $[-1, 1]$. The derivatives described in Eqs. (6) and (13) depend on the scaling function. Thus, both equations are modified to account for the scaling by application of the chain rule for the respective equation incorporating the scaling function in accordance with [17].

4.2.3. Design of experiment (DoE) for the training of surrogate models with parametric P-Boxes as input quantities

The identification of the corners of the hypercube in $\mathcal{U}$ is dependent on the type of uncertainty evaluated. For interval-valued uncertainties, as described in Section 3.2, the bounds of the corresponding quantities are determined straightforwardly by the interval

of the associated parameter itself. If fuzzy-values as in [35] are concerned, they are discretizable as α -levels, and the minimal and maximal boundaries of the α -levels are the boundaries of the hypercube.

If stochastic quantities, as described in Section 3.1, are considered in U , finding intervals for the associated quantity, in which samples are drawn, is not straightforward. Continuous probability distributions may have an unconstrained domain of definition at either one or both of the sides. Thus, it is proposed to use confidence intervals $\mathcal{A}_\beta$ as introduced in Section 3.1. For that, a confidence level β needs to be chosen. The confidence interval must be selected in a way that as few points as possible are drawn outside of the boundaries of the confidence intervals in the uncertainty quantification. Thus, $0 < \beta \ll 1$ is used. If the confidence level is set appropriately, the PANN will behave well even when samples are drawn outside of the selected confidence interval, because the samples drawn outside will likely be near the borders of the confidence interval. Several sets $\mathcal{A}_\beta$ fulfill the condition (see Eq. (16)) of a confidence interval for a specific β . Based on the assumption of quantiles, as in Section 3.1, the bounds of the confidence intervals are determined, which entails a set of boundaries for the random number, thereby providing a sample of the respective CDF with a probability β . This assumption is independent of the form of the CDF.

In the case of random fields, at each observation point in $\mathbf{X}$, marginal distributions are underlying. If the random field is stationary, the same marginal distribution underlies at each discretized point, thus, only one confidence interval for this marginal must be selected. Otherwise, the maximum and minimum boundaries of all the marginals describing the same parameter are utilized to ensure a minimal confidence.

In case of parametric p-boxes, as introduced in Section 3.3, the confidence intervals described in Section 3.1 are dependent on θ by $\mathcal{A}_\beta(\theta)$. Therefore, Eq. (16) is now expanded as

$$P(\mathcal{A}_\beta(\theta)) \geq 1 - \beta, \quad \beta \in (0, 1), \text{ which holds for all } \theta \in \Theta^i, \quad (29)$$

hence, for all parameter vectors θ of the interval input space Θ^i . For a fixed value β , a set of confidence intervals is obtained from a p-box. To define the confidence interval bounds, the domain described in Eq. (29) is evaluated after appropriate bounds for the hypercube as

$$\mathcal{A}_{\beta, \max}^{\text{pb}}(\Theta^i) = \{\omega \in \Omega \mid \min_{\theta \in \Theta} a_l(\theta) \leq V^T(\omega) \leq \max_{\theta \in \Theta} a_u(\theta)\}. \quad (30)$$

The set described by Eq. (30) is an interval, capturing all possible sampling points of the p-boxes with a probability of at least β . If Eq. (29) is applied on Eq. (30) as

$$1 - \beta \leq P(\mathcal{A}_{\beta, \max}^{\text{pb}}(\Theta^i)) \leq 1, \quad (31)$$

the probability of drawing a sample point from the swarm of distributions that build the p-box is bound by $[1 - \beta, 1]$, as stated in Eq. (31) and $P(\mathcal{A}_{\beta, \max}^{\text{pb}}(\Theta^i))$ itself is an interval-valued probability.

The hypercube itself is then constructed as CARTESIAN product of all the maximal and minimal interval bounds of all U^i , $\mathcal{A}_\beta$, $\mathcal{A}_{\beta, \max}^{\text{pb}}(\Theta^i)$ for each parameter in U .

4.2.4. DoE For the mechanical domain

To train the PANN, the mechanical domain $\mathcal{F}$ and the domain of uncertain parameters U are separated sampling spaces as described in Section 4.2.2 and, in this contribution, explored by LHS, as described in Section 4.2. For the mechanical domain, it is assumed that all mechanical states are relevant for every RVE. Thus, one hypercube is identified for all RVEs. A specific LHS is carried out for each sample $\mathbf{u} \in U$. To find the boundaries of the hypercube, knowledge about the induced strain is required to draw samples in $\mathcal{F}$. In the contribution at hand, the approach presented in [27] is followed, where a reasonable bound for $\mathcal{F}$ is chosen based on knowledge of the material behavior. Different approaches, as in [17], based on the boundary value problem, may be used. Because it is unclear how large the actual maximal strain is, based on the uncertain macroscopic boundary conditions, large bounds are selected. If the bounds of $\mathcal{F}$ are set inappropriately and the PANN needs to extrapolate the stiffness for deformations far away from the training domain, the auto-differentiation of the PANN, which is needed to reconstruct $\bar{\mathbf{C}}^{\text{NN}}$, will be numerically unstable. This is mitigated by setting the bounds of $\mathcal{F}$ appropriately.

5. Numerical examples

In order to demonstrate the applicability of the presented methods, two examples are evaluated. In the first example, it is shown that a parametrized constitutive model characterizing hyperelasticity can be surrogated by the proposed PANN.

In the second example, the complete proposed framework of multiscale polymorphic uncertainty quantification based on PANNs is demonstrated. A numerical reference solution is not available with a feasible computational effort for this second numerical example, but the investigated constitutive model of the first example can be analytically evaluated. Hence, the advantage of an available reference solution in the first example is facilitated to validate the proposed methods.

The PANN is implemented in TensorFlow [49]. It is highlighted that a hyperparameter optimization is incorporated (see [50]) for the construction of the PANN structure, in contrast to fixed structures as in [13,28]. For the hyperparameter optimization, the training data set is split into 90% training data and 10% validation data. Thereby, the training of the PANN is started multiple times, in both examples 5 per optimization step, to ensure independence of the initial weights. Also, 50 different architectures are tested in the hyperparameter optimization. The obtained PANN is incorporated into the FE scheme FEAP [51] as material model. Solutions are then obtained by evaluating the uncertainty within the FE scheme using FEAP as a finite element solver. In order to calculate correlated

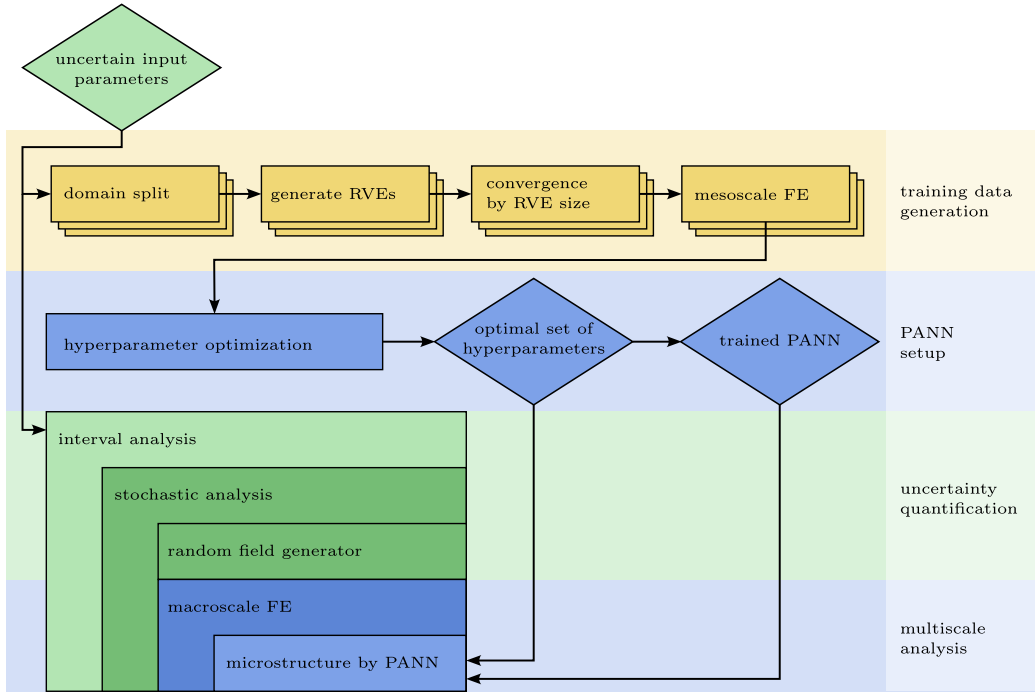


Fig. 5. Flowchart of the algorithm applied in the examples.

random samples based on a given correlation structure, the KARHUNEN-LOÉVE expansion [38] is utilized. In Fig. 5, a flowchart of the final framework, as used in Section 5.1 and Section 5.2, is presented. There, key aspects of the algorithm are highlighted.

As quantity of interest of the FE simulation, the maximal value

$$\sigma_{\text{char}} = \max(\max_{\mathbf{x} \in B_i} |\sigma_i(\mathbf{x})|) \quad (32)$$

of all principal stresses $\sigma_1, \dots, \sigma_3$ at each of the nodes $\mathbf{x}$ in the body B_i of the FE-simulation is used. This ensures the most extreme behavior of the examples is captured and enables the PANN to be capable of approximating the relevant states for a risk assessment in rare events. To obtain the principal stresses, the true stresses σ , which are obtained by the FE-simulation, are evaluated by their eigenvalues σ_i .

5.1. Plate with hole – Analytical solution based on NEO-HOOKE – Numerical example I

Setup. Within this example, two solutions are compared with each other. Because a reference solution based on an FE² scheme is numerically too expensive to conduct, a parametrized strain energy density based on uncertain parameters is exemplary studied on a parametrized, isotropic material description, the modified NEO-HOOKEan law

$$\mathbf{S} = \mathbf{F}^{-1} \cdot \left(\kappa \ln[J] \mathbf{F}^{-T} + \eta \left(J^{-\frac{2}{3}} \mathbf{F} - \frac{\text{tr}(J^{-\frac{2}{3}} \cdot \mathbf{C})}{3} \mathbf{F}^{-T} \right) \right), \quad (33)$$

with $\kappa = \frac{E}{3 \cdot (1-2\nu)}$ and $\eta = \frac{E}{2 \cdot (1+\nu)}$.

The material parametrization of the NEO-HOOKEan material is formulated through E and ν . The YOUNG's modulus E is described by an interval probability-based random field E^{iprf} . Thereby, the marginal distribution of the random field is described by a GAUSSIAN normal distribution. E is the variable of the events of the CDF (see Section 3.1). The parameters of the distribution are $\mu_E^i = [2.8, 3.2] \cdot 10^4$ in MPa and $\sigma_E = 10^3$ MPa. Because μ_E^i is an interval-valued parameter of the CDF, the resulting swarm of CDFs describes E^{iprf} . The correlation structure of E^{iprf} is based on a *squared exponential* autocorrelation function as described in [39]. The autocorrelation length is assumed by $l_{\text{corr},E} = 1$ m. Additionally, a random field ν^{rf} is used to model the uncertainty of ν . Here, the marginal CDF is modeled with a GAUSSIAN normal distribution. The parameters of the marginal are $\mu_\nu = 0.3$, $\sigma_\nu = 0.03$ and a squared exponential autocorrelation function with $l_{\text{corr},\nu} = 2$ m. The last uncertain parameter is the displacement. It is a random variable, following the parametrization of *logarithmic normal distributions* with $\mu_U = -0.12$ m, $\sigma_U = 0.05$ m and $x_0 = -0.2$ m according to [52].

The macroscopic structure investigated is presented in Fig. 6, with the parameters $l = 8$ m, $b = 2$ m and $r = 2$ m. The mesh consists of 8335 linear tetrahedral elements. The plate is evaluated for the quantity of interest σ_{char} , see Eq. (32). Training data is acquired by the combined LH-sampling as described in Section 4.2.2. The obtained bounds of the hypercube per uncertain parameter and their respective confidence levels are shown in Table 1. The mechanical domain is evaluated in a range of $[-0.2, 0.2]$ for each entry of

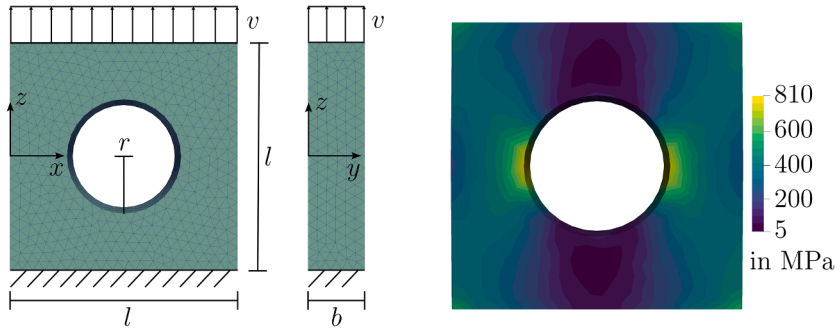


Fig. 6. Geometry (left) and resulting magnitude of principle stresses for an exemplary realization of the plate with hole (right).

Table 1

Design of experiment bounds of the probability-based uncertain parameters.

parameter	lower bound a_l	upper bound a_u	confidence β
E^{pref} in MPa	$2.5 \cdot 10^4$	$3.5 \cdot 10^4$	2.8×10^{-3}
ν^{rf} in $[-]$	0.21	0.39	2.8×10^{-3}

Table 2

Structure of the PANN obtained from the hyperparameter optimization.

layer	number of neurons	activation function
Layer 1	5	softplus
Layer 2	175	softplus
Layer 3	175	softplus
Layer 4	1	linear

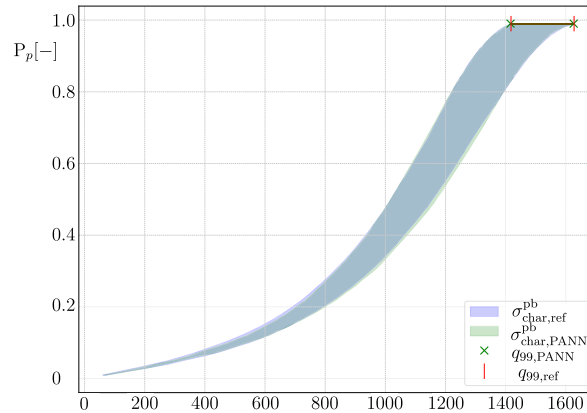


Fig. 7. Result of the multiscale uncertainty quantifications as p-box for the quantity of interest $\sigma_{\text{char}}^{\text{pb}}$ – PANN-based solution $\sigma_{\text{char}}^{\text{pb}}_{\text{PANN}}$ in comparison with reference solution $\sigma_{\text{char}}^{\text{pb}}_{\text{ref}}$, discretized by quantiles.

$\mathbf{H}$ and $\mathbf{F} = \mathbf{H} + \mathbf{1}$. Within the hypercube of uncertain parameters, 50 samples are drawn. For each LHS in the mechanical space, the sampling is constructed with 100 points.

Results. As an information reducing measure (see [32]) for the stochastic analysis, the 99% quantile $q_{99, \sigma_{\text{char}}}$ of the resulting CDF is evaluated. As the characteristic value, σ_{char} is utilized Table 2.

The results are presented in Fig. 7. The example is evaluated by an interval analysis, which evaluates a MONTE-CARLO simulation that quantifies the stochastic response of the system. As an optimization algorithm for the interval analysis, the “evolutionary strategy” [53] is utilized. Thereby, the 99%-quantile q_{99} of σ_{char} is used as optimization criterion for the interval analysis. The MONTE-CARLO simulation is conducted by drawing 10^4 samples, evaluating the resulting q_{99} and then, redrawing $2 \cdot 10^3$ samples until q_{99} does not change more than 0.5% within the last 5 steps.

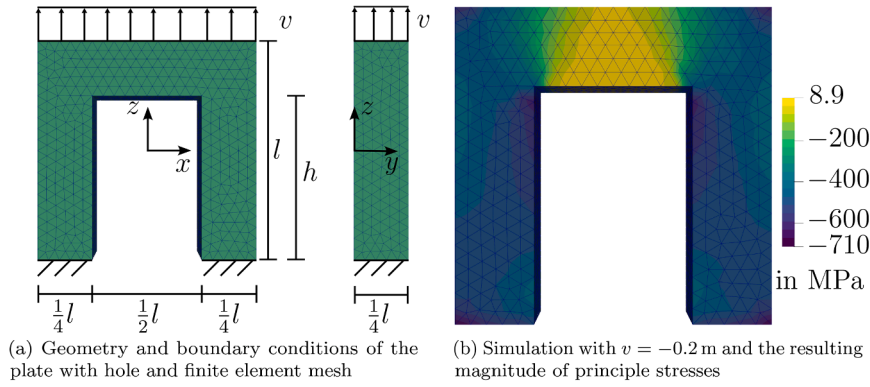


Fig. 8. Investigated FE model.

Table 3

Parameters for the setup and evaluated parameter domain.

parameter	value	lower bound a_l	upper bound a_u	confidence β
v_f	$v_f^{\text{pb,rf}}$ in %	10	40	2.8×10^{-3}
E_{matrix}	E_{rf} in MPa	16253	17375	2.8×10^{-3}
h	h^l in m	$0.5 \cdot l$	$0.8 \cdot l$	–
$E_{\text{inclusion}}$	5×10^4 MPa	–	–	–
l_{RVE}	60 mm	–	–	–
l	8 m	–	–	–
v	v^f in m	–	–	–

The PANN-structure obtained from the hyperparameter optimization is shown in Table 4. The results presented in Fig. 7 show the p-boxes of the reference solution and the PANN solution, discretized by the respective quantiles. The point at the obtained results in the input domain is received at the bounds of the input interval as $[2.8 \cdot 10^4, 3.2 \cdot 10^4]$ for both simulations. The relative error $\epsilon_{\text{rel}} = \frac{|q_{99,\text{ref}} - q_{99,\text{PANN}}|}{q_{99,\text{ref}}}$ for both interval bounds between reference solution and PANN-based solution is 0.1% for the left bound and 0.07% for the right bound.

The fact that the optimized q_{99} of both solutions are nearly overlapping shows the effectiveness of the method. The results are shown in Fig. 7 and depicted by the intervals at the top right corner. The other boundaries of the p-boxes in the result space for the PANN $\sigma_{\text{char,PANN}}^{\text{pb}}$ and the reference solution $\sigma_{\text{char,ref}}^{\text{pb}}$ are obtained by evaluating several quantiles of the sampling points of the MONTE-CARLO simulation at the solution points of the optimization algorithm. They are not a separate criterion for the optimization. Due to stochastic variability, the resulting boundaries do not fully align. But they display similar characteristics. The effective error for the quantiles of the p-box bounds cannot be interpreted as relative error between both methods, in contrast to the error between $q_{99,\text{ref}}$ and $q_{99,\text{PANN}}$.

5.2. Complete framework of multiscale polymorphic uncertainty quantification with PANNs – Numerical example II

Setup. In this example, the complete uncertainty quantification as introduced and discussed in Section 4 is conducted based on the RVE introduced in [54]. The macroscopic quantity of interest is $\bar{S}$ (see Section 2.2), because the PANN is trained based on this quantity (see Section 2.5). The material with ellipsoidal inclusions described by the RVE is isotropic, if $\bar{S}$ converges against its volume average. Thus, the first three invariants are utilized in network training. The volume fraction v_f and the YOUNG's modulus E_{matrix} of the matrix are assumed to be uncertain. Thereby, the volume fraction is an interval probability-based random field v_f^{iprf} . Its marginal CDF is assumed as GAUSSIAN normal distribution with n interval valued mean $\mu_{v_f}^i = [16\%, 34\%]$ and a standard deviation of $\sigma_{v_f} = 2\%$. A squared exponential autocorrelation function is assumed to describe the spatial relation between different volume fractions on the macroscale, with a characteristic length of $l_{\text{corr},v_f} = 2$ m.

Assumptions about the uncertainty are based on the parameters used in [6,54]. An exemplary, meshed realization of the structure and its evaluation with periodic boundary conditions is shown in Fig. 9. The mesoscopic value of the YOUNG's modulus of the cement matrix is considered uncertain on the macroscale but deterministic at the mesoscale of the respective realization. It is described by a random field on the macroscale $E_{\text{matrix}}^{\text{rf}}$. Its marginal CDF is assumed as GAUSSIAN distribution, with parameters $\mu_{E_{\text{matrix}}} = 16814$ MPa and $\sigma_{E_{\text{matrix}}} = 187$ MPa, the autocorrelation structure is assumed as a squared exponential autocorrelation function with $l_{\text{corr,matrix}} = 1$ m.

For each RVE, the domain is meshed with linear tetrahedral elements. The amount of tetrahedral elements in each RVE is based on the realization of the RVE geometry. For the sampling in the domain of uncertain parameters, 50 samples are drawn and then checked for convergence of the quantity of interest $\bar{S}$. As criterion for the change of $\bar{S}$, the tensor norm $\|\bar{S}\|$ is evaluated by 1000 samples, with a prescribed deformation of $\bar{\mathbf{F}} = \mathbf{1} + \bar{\mathbf{H}}$, where each entry of $\bar{\mathbf{H}}$ equals 0.1. Thus, a total of 5×10^4 training data points

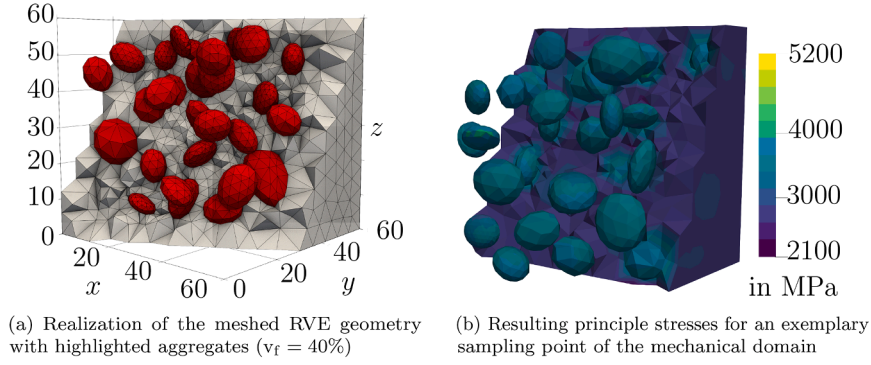


Fig. 9. Exemplary structure of the RVE introduced in [54] and resulting uncertainty.

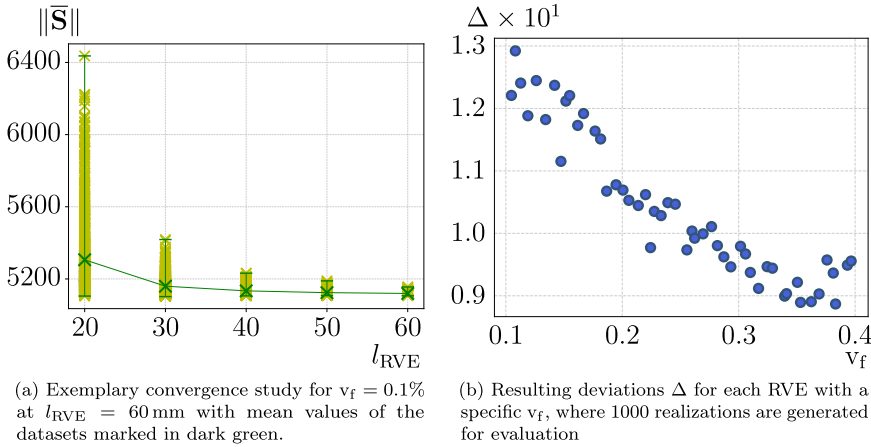


Fig. 10. Exemplary convergence test and convergence analysis of all training data in U .

is generated, and an equal amount of RVE evaluations is conducted with a total amount of 43 CPU hours. As a criterion of change, the standard deviation of the obtained sample points for each RVE $\Delta = \frac{E(\|\bar{S}\|) - \sqrt{\text{VAR}(\|\bar{S}\|)}}{E(\|\bar{S}\|)}$ in [%], is used. An RVE, where $\Delta \leq 0.5\%$ is assumed to be converged against its effective macroscopic property. For each RVE sample, 1000 samples are drawn in the mechanical space. In Fig. 10, the results of the convergence test are displayed. An exemplary convergence study is carried out to ensure proper convergence behavior of the RVEs under periodic boundary conditions. Then, the results for the convergence of all sampled RVEs are depicted in relation to the volume fraction.

As a macroscopic structure, the system depicted in Fig. 8 is evaluated. The prescribed displacement v is assumed as a stochastic quantity v^s , with a CDF based on a *logarithmic normal distribution*, as defined and parametrized in [52]. The parameters are assumed as $\mu_U = -0.1$ m, $\sigma_U = 0.15$ m and $x_0 = -0.6$ m. The height of the exclusion is parametrized with $h^i \cdot l$. The scale of the height is an interval quantity in the range of $h^i = [0.5 \cdot l, 0.8 \cdot l]$ describing an imperfection of the system. Thus, for every h drawn by the optimization scheme, a new structure is meshed, and the macroscale boundary value problem changes. The FE-mesh consists of linear tetrahedral elements. Their number and position depend on the realization of h and the results of the meshing, ranging between 6000 and 10000 finite elements. Thus, for each step in the optimization, a new mesh is generated, and the random field is evaluated with new points. In Table 3, all parameters are listed and, if applicable, the respective range of evaluation and corresponding confidence is presented.

Results. As initial VE-size, a length of $l_{RVE} = 60$ mm is chosen, because after a pre-evaluation, those geometries displayed a low number of finite elements. By testing the convergence, it is found that the initially chosen length, all RVEs sampled in U , converge under the condition of periodic boundary conditions. As a solution scheme, the same scheme as in Section 5.1 is utilized. The PANN-structure obtained from the hyperparameter optimization is shown in Table 4. As error measurements, the same measurements are used. The results are presented in Fig. 11.

Within the evaluation of the framework with an FE^2 approach as basic solution, at each GAUSSIAN point, 6000 – 10000 RVEs are evaluated multiple times in the macroscopic NEWTON-RAPHSON scheme until their solution converges. One evaluation of the NEWTON-RAPHSON scheme of an exemplary RVE on a workstation with an Intel Core i5-10400×12 CPU and without acceleration by a graphics card, takes 203.32 s. In contrast to that, the evaluation of the PANN takes around 0.0005 s. Therefore, even one evaluation of the basic solution with a PANN is by magnitudes faster than the FE simulation of a single RVE, which just reflects a single integration point of a macrostructure. Also, in the whole uncertainty quantification, 566000 basic solutions are evaluated. The total amount of

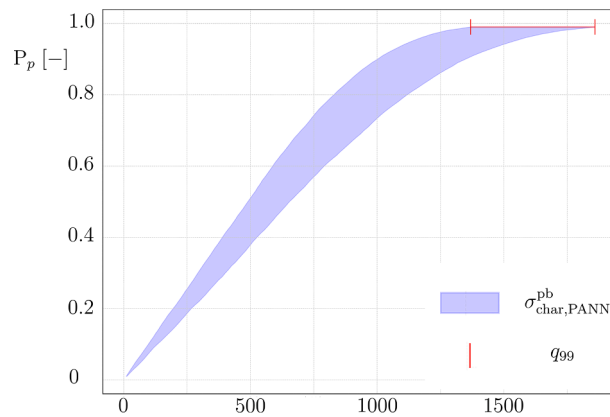


Fig. 11. Result of the multiscale uncertainty quantifications as p-box for the quantity of interest $\sigma_{\text{char}}^{\text{pb}}$ – PANN-based solution $\sigma_{\text{char,PANN}}^{\text{pb}}$, discretized by quantiles.

Table 4

Structure of the PANN obtained from the hyperparameter optimization.

layer	number of neurons	activation function
Layer 1	5	softplus
Layer 2	170	softplus
Layer 3	55	softplus
Layer 4	1	linear

time for the hyperparameter optimization in serial TensorFlow is 48 h, and the training data generation about 43 h. Even with this amount of time needed for setting up the PANN, the proposed framework is far more efficient than an uncertainty quantification of the same problem with an FE^2 approach as a basic solution.

The resulting interval-valued quantile is obtained with $q_{99} = [1368.8 \text{ MPa}, 1859.7 \text{ MPa}]$. Those values are evaluated near the bounds of the input interval at $[18\%, 0.8 \cdot I]$ and $[33.9\%, 0.5 \cdot I]$. This is reasonable, because $E_{\text{matrix}} < E_{\text{inclusion}}$ and a higher volume fraction means more inclusions and thus, the material is stiffer. The resulting p-box at the solution points of the optimization is depicted by $\sigma_{\text{char}}^{\text{pb}}$. It is built through evaluation of quantiles of the data points at the results of the optimization algorithm. This example demonstrates the full solution scheme and highlights the possibility of conducting the uncertainty quantification independent of the discretization of the macroscopic structure. Thus, even geometrical imperfections are considerable. Also, it highlights the fact that several different uncertainty models (aleatoric, epistemic, and polymorphic models) are evaluated simultaneously.

6. Conclusion

In this contribution, the behavior of mesostructures is simulated to use it as a constitutive law for macrostructures. Thereby, polymorphic uncertainty is considered on the macro- and mesoscale to assess the safety of the macroscopic structures. The fundamental aspects of polymorphic uncertainty in the context of multi-scale analysis are discussed. The propagation of mesoscale uncertainty through homogenization is discussed. Because of the high numerical effort, a homogenization technique is introduced into the uncertainty quantification, the consideration of uncertainty in homogenization is enabled through PANNs with uncertain input parameters. For the PANNs with uncertain input parameters, an efficient sampling scheme is developed, which allows the reduction of polymorphic input domains to a parametrized hypercube, independent of the type of uncertainty. For that, the concept of confidence intervals for p-boxes is introduced. Subsequently, the split between the mechanical and uncertain input domain is proposed, and its efficiency is discussed.

The surrogate by the PANN allows an efficient sampling for the polymorphic uncertainty quantification scheme. Because of the sampling-based nature of the uncertainty quantification algorithms, the reduction of the uncertainty to a parameter domain where those uncertainty analyses draw samples enables all types of uncertainties to be considered.

The applicability of the proposed methods is demonstrated by numerical examples of two engineering structures. Therein, interval, stochastic, and p-box input quantities are considered on different scales (meso- and macroscale). The first example demonstrates the effectiveness of the proposed method by comparing an analytical constitutive law with a PANN. The PANN is used as a surrogate for that constitutive law, and the results of both uncertainty quantifications are compared. In the second example, the proposed methods are applied to an RVE with uncertain structural descriptors. The possibility of conducting the uncertainty quantification based on geometrical imperfections of the macroscopic structure and the simultaneous consideration of several uncertainty models is demonstrated.

Because the material symmetries need to be known and not changing based on the parametrization, the proposed approach is restricted in that regard, and a formulation with less physical information has to be adapted. The substance of further research is the possibility of using arbitrary material symmetries without loss of physical information in the network. Also, the consideration of stochastic volume elements and non-converged RVEs leads to a stochastic response of the energy density, which is not solvable by this approach. Further research is required to extend the solution scheme in order to incorporate that type of uncertainty into the solution algorithm.

CRedit authorship contribution statement

Felix Harazin: Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization; **Jakob Platen:** Writing – review & editing, Supervision, Software, Methodology; **F. Niklas Schietzold:** Writing – review & editing, Methodology, Conceptualization; **Wolfgang Graf:** Writing – review & editing, Supervision; **Michael Kaliske:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Demonstrated results and research activities arise from the following projects of the DFG (Deutsche Forschungsgemeinschaft). Research Training Group GRK 2868 “Data-Driven Design of Resilient Metamaterials” (D3) - Project No. 493401063 as well as the Priority Program SPP 1886 with the corresponding grant numbers GR 1504/10, GR 1504/11, KA 1163/36, KA 1164/37 and KA 1164/38 and Project No. 414936990 under grant number KA 1163/46-2. The authors gratefully acknowledge the computing time made available to them on the high-performance computer at the NHR Center of TU Dresden. This center is jointly supported by the Federal Ministry of Research, Technology and Space of Germany and the state governments participating in the NHR (www.nhr-verein.de/unsere-partner).

References

- [1] R. Bostanabad, B. Liang, J. Gao, W.K. Liu, J. Cao, D. Zeng, X. Su, H. Xu, Y. Li, W. Chen, Uncertainty quantification in multiscale simulation of woven fiber composites, *Comput. Methods Appl. Mech. Eng.* 338 (2018) 506–532. <https://doi.org/10.1016/j.cma.2018.04.024>
- [2] M. Komeili, A.S. Milani, The effect of meso-level uncertainties on the mechanical response of woven fabric composites under axial loading, *Comput. Struct.* 90–91 (2012) 163–171. <https://doi.org/10.1016/j.compstruc.2011.09.001>
- [3] S.L. Omairey, P.D. Dunning, S. Sriramula, Influence of micro-scale uncertainties on the reliability of fibre-matrix composites, *Compos. Struct.* 203 (2018) 204–216. <https://doi.org/10.1016/j.compstruct.2018.07.026>
- [4] M.J. Bogdanor, C. Oskay, S.B. Clay, Multiscale modeling of failure in composites under model parameter uncertainty, *Comput. Mech.* 56 (2015) 389–404. <https://doi.org/10.1007/s00466-015-1177-7>
- [5] A. Clément, C. Soize, J. Yvonnet, Uncertainty quantification in computational stochastic multiscale analysis of nonlinear elastic materials, *Comput. Methods Appl. Mech. Eng.* 254 (2013) 61–82. <https://doi.org/10.1016/j.cma.2012.10.016>
- [6] F. Leichenring, *Ausgewählte numerische Analyse- und Entwurfsmethoden mit polymorph unscharfen Datenmodellierungen*, Dissertationsschrift, Technische Universität Dresden, Publications of the Institute for Structural Analysis, Issue 42, 2021.
- [7] B.A. Le, J. Yvonnet, Q.C. He, Computational homogenization of nonlinear elastic materials using neural networks, *Int. J. Numer. Methods Eng.* 104 (2015) 1061–1084. <https://doi.org/10.1002/nme.4953>
- [8] Z. Liu, M.A. Bessa, W.K. Liu, Self-consistent clustering analysis: an efficient multi-scale scheme for inelastic heterogeneous materials, *Comput. Methods Appl. Mech. Eng.* 306 (2016) 319–341. <https://doi.org/10.1016/j.cma.2016.04.004>
- [9] D. Pivovarov, V. Hahn, P. Steinmann, K. Willner, S. Leyendecker, Fuzzy dynamics of multibody systems with polymorphic uncertainty in the material microstructure, *Comput. Mech.* 64 (2019) 1601–1619. <https://doi.org/10.1007/s00466-019-01737-9>
- [10] D. Pivovarov, T. Oberleiter, K. Willner, P. Steinmann, Fuzzy-stochastic FEM-based homogenization framework for materials with polymorphic uncertainties in the microstructure, *Int. J. Numer. Methods Eng.* 116 (2018) 633–660. <https://doi.org/10.1002/nme.5947>
- [11] Z. Zheng, U. Nackenhorst, Efficient uncertainty propagation for stochastic multiscale linear elasticity, *Comput. Methods Appl. Mech. Eng.* 428 (2024) 117085. <https://doi.org/10.1016/j.cma.2024.117085>
- [12] S. Zschocke, F. Leichenring, W. Graf, M. Kaliske, A concept for data-driven computational mechanics in the presence of polymorphic uncertain properties, *Eng. Struct.* 267 (2022) 114672. <https://doi.org/10.1016/j.engstruct.2022.114672>
- [13] L. Linden, D.K. Klein, K.A. Kalina, J. Brummund, O. Weeger, M. Kästner, Neural networks meet hyperelasticity: a guide to enforcing physics, *J. Mech. Phys. Solids* 179 (2023) 105363. <https://doi.org/10.1016/j.jmps.2023.105363>
- [14] R. Fleischhauer, *A Boundary Value Driven Approach to Computational Homogenization*, Dissertationsschrift, Technische Universität Dresden, Publications of the Institute for Structural Analysis, Issue 28, 2016.
- [15] G.A. Holzapfel, *Nonlinear Solid Mechanics – A Continuum Approach for Engineering*, Wiley, Chichester, 2000.
- [16] A. Ricker, P. Wriggers, Systematic fitting and comparison of hyperelastic continuum models for elastomers, *Arch. Comput. Methods Eng.* 30 (2023) 2257–2288. <https://doi.org/10.15488/14634>
- [17] K.A. Kalina, L. Linden, J. Brummund, M. Kästner, FE^{ANN} : An efficient data-driven multiscale approach based on physics-constrained neural networks and automated data mining, *Comput. Mech.* 71 (2023) 827–851. <https://doi.org/10.1007/s00466-022-02260-0>

- [18] M. Ostoja-Starzewski, Material spatial randomness: from statistical to representative volume element, *Probab. Eng. Mech.* 21 (2006) 112–132. <https://doi.org/10.1016/j.probenmech.2005.07.007>
- [19] S. Torquato, et al., *Random Heterogeneous Materials: Microstructure and Macroscopic Properties*, Springer Science & Business Media, New York, 2002.
- [20] R. Hill, Elastic properties of reinforced solids: some theoretical principles, *J. Mech. Phys. Solids* 11 (1963) 357–372. [https://doi.org/10.1016/0022-5096\(63\)90036-X](https://doi.org/10.1016/0022-5096(63)90036-X)
- [21] V. Kouznetsova, *Computational homogenization for the multi-scale analysis of multi-phase materials*, Technische Universiteit Eindhoven, 2002.
- [22] S. Torquato, Hyperuniform states of matter, *Phys. Rep.* 745 (2018) 1–95. <https://doi.org/10.1016/j.physrep.2018.03.001>
- [23] C. Miehe, Computational micro-to-macro transitions for discretized micro-structures of heterogeneous materials at finite strains based on the minimization of averaged incremental energy, *Comput. Methods Appl. Mech. Eng.* 192 (2003) 559–591. [https://doi.org/10.1016/S0045-7825\(02\)00564-9](https://doi.org/10.1016/S0045-7825(02)00564-9)
- [24] Z.F. Khisaeva, M. Ostoja-Starzewski, On the size of RVE in finite elasticity of random composites, *J. Elast.* 85 (2006) 153–173. <https://doi.org/10.1007/s10659-006-9076-y>
- [25] K. Terada, M. Hori, T. Kyoya, N. Kikuchi, Simulation of the multi-scale convergence in computational homogenization approaches, *Int. J. Solids Struct.* 37 (2000) 2285–2311. [https://doi.org/10.1016/S0020-7683\(98\)00341-2](https://doi.org/10.1016/S0020-7683(98)00341-2)
- [26] J. Platen, J. Storm, S. Bosbach, M. Claßen, M. Kaliske, The microlayer model: a novel analytical homogenisation scheme for materials with rigid particles and deformable matrix – applied to simulate concrete, *Comput. Struct.* 293 (2024) 107258. <https://doi.org/10.1016/j.compstruc.2023.107258>
- [27] J.P. Stöcker, J. Platen, M. Kaliske, Introduction of a recurrent neural network constitutive description within an implicit gradient enhanced damage framework, *Comput. Struct.* 289 (2023) 107162. <https://doi.org/10.1016/j.compstruc.2023.107162>
- [28] K.A. Kalina, L. Linden, J. Brummund, P. Metsch, M. Kästner, Automated constitutive modeling of isotropic hyperelasticity based on artificial neural networks, *Comput. Mech.* 69 (2022) 213–232. <https://doi.org/10.1007/s00466-021-02090-6>
- [29] L. Fahrmeir, C. Heumann, R. Künstler, I. Pigeot, G. Tutz, *Statistik: Der Weg zur Datenanalyse*, Springer Spektrum, Heidelberg, 2024. <https://doi.org/10.1007/978-3-662-67526-7>
- [30] N. Henze, *Stochastik: eine Einführung mit Grundzügen der Maßtheorie*, Springer Spektrum, Heidelberg, 2019. <https://doi.org/10.1007/978-3-662-59563-3>
- [31] M. Götz, *Numerische Entwurfsmethoden Unter Berücksichtigung Polymorpher Unschärfe*, Dissertationsschrift, Technische Universität Dresden, Publications of the Institute for Structural Analysis, Issue 32, 2017.
- [32] F.N. Schietzold, F. Leichsenring, M. Götz, W. Graf, M. Kaliske, Robustness versus performance - Nested inheritance of objectives in optimization with polymorphic uncertain parameters, *Adv. Eng. Software* 156 (2021) 102932. <https://doi.org/10.1016/j.advengsoft.2020.102932>
- [33] B. Möller, M. Beer, *Fuzzy randomness: uncertainty in civil engineering and computational mechanics*, Springer, Heidelberg, 2004.
- [34] M. Beer, S. Ferson, V. Kreinovich, Imprecise probabilities in engineering analyses, *Mech. Syst. Signal. Process.* 37 (2013) 4–29. <https://doi.org/10.1016/j.ymssp.2013.01.024>
- [35] W. Graf, M. Götz, M. Kaliske, Analysis of dynamical processes under consideration of polymorphic uncertainty, *Struct. Saf.* 52 (2015) 194–201. <https://doi.org/10.1016/j.strusafe.2014.09.003>
- [36] R. Schöbi, B. Sudret, Structural reliability analysis for p-boxes using multi-level meta-models, *Probab. Eng. Mech.* 48 (2017) 27–38. <https://doi.org/10.1016/j.probenmech.2017.04.001>
- [37] F. Beichelt, *Stochastische Prozesse für Ingenieure*, B.G. Teubner, Stuttgart, 1997.
- [38] B. Sudret, A.D. Kiureghian, *Stochastic Finite Element Methods and Reliability - A State-of-the-Art Report*, Structural Engineering, Mechanics and Materials Department of Civil & Environmental Engineering, University of California, Berkeley, Technical Report No. UCB/SEMM-2000/08, 2000.
- [39] F.N. Schietzold, W. Graf, M. Kaliske, Multi-Objective optimization of tree trunk axes in glulam beam design considering fuzzy probability-Based random fields, *ASCE-ASME J. Risk and Uncert. Eng. Syst. Part B: Mech. Eng.* 7 (2021) 020913. <https://doi.org/10.1115/1.4050370>
- [40] F.N. Schietzold, A. Schmidt, M.M. Dannert, A. Fau, R.M.N. Fleury, W. Graf, M. Kaliske, C. Könke, T. Lahmer, U. Nackenhorst, Development of fuzzy probability based random fields for the numerical structural design, *GAMM-Mitteilungen* 42 (2019) e201900004. <https://doi.org/10.1002/gamm.201900004>
- [41] M. Vořechovský, Simulation of simply cross correlated random fields by series expansion methods, *Struct. Saf.* 30 (2008) 337–363. <https://doi.org/10.1016/j.strusafe.2007.05.002>
- [42] J.N. Fuhg, N. Bouklas, On physics-informed data-driven isotropic and anisotropic constitutive models through probabilistic machine learning and space-filling sampling, *Comput. Methods Appl. Mech. Eng.* 394 (2022) 114915. <https://doi.org/10.1016/j.cma.2022.114915>
- [43] J. Ghaboussi, J.H. Garrett, X. Wu, Knowledge-Based modeling of material behavior with neural networks, *J. Eng. Mech.* 117 (1991) 132–153. [https://doi.org/10.1061/\(ASCE\)0733-9399\(1991\)117:1\(132\)](https://doi.org/10.1061/(ASCE)0733-9399(1991)117:1(132))
- [44] Y.M.A. Hashash, S. Jung, J. Ghaboussi, Numerical implementation of a neural network based material model in finite element analysis, *Int. J. Numer. Methods Eng.* 59 (2004) 989–1005. <https://doi.org/10.1002/nme.905>
- [45] A.A. Jadoon, K.A. Kalina, M.K. Rausch, R. Jones, J.N. Fuhg, Inverse design of anisotropic microstructures using physics-augmented neural networks, 2024. [arXiv:2412.13370](https://arxiv.org/abs/2412.13370).
- [46] D.K. Klein, F.J. Roth, I. Valizadeh, O. Weeger, Parametrized polyconvex hyperelasticity with physics-augmented neural networks, *Data-Centric Eng.* 4 (2023) e25. <https://doi.org/10.1017/dce.2023.21>
- [47] J. Yvonnet, D. Gonzalez, Q.C. He, Numerically explicit potentials for the homogenization of nonlinear elastic heterogeneous materials, *Comput. Methods Appl. Mech. Eng.* 198 (2009) 2723–2737. <https://doi.org/10.1016/j.cma.2009.03.017>
- [48] M.D. McKay, R.J. Beckman, W.J. Conover, A comparison of three methods for selecting values of input variables in the analysis of output from a computer code, *Technometrics* 21 (1979) 55–61. <https://doi.org/10.1080/00401706.2000.10485979>
- [49] M. Abadi, A. Agarwal, P. Barham, E. Brevdo, Z. Chen, C. Citro, et al., TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems, 2015. <https://www.tensorflow.org/>.
- [50] T. O'Malley, E. Bursztajn, J. Long, F. Chollet, H. Jin, L. Invernizzi, et al., KerasTuner, 2019. <https://github.com/keras-team/keras-tuner>.
- [51] R.L. Taylor, FEAP - Finite Element Analysis Program, 2014. <http://www.ce.berkeley.edu/feap>.
- [52] G. Spaethe, *Die Sicherheit tragender Baukonstruktionen*, Springer, Vienna, 2nd Edition, 1992.
- [53] J. Blank, K. Deb, Pymoo: multi-Objective optimization in python, *IEEE Access* 8 (2020) 89497–89509. <https://doi.org/10.1109/ACCESS.2020.2990567>
- [54] S. Eckardt, C. Könke, Adaptive damage simulation of concrete using heterogeneous multiscale models, *J. Algor. Computat. Technol.* 2 (2008) 275–298. <https://doi.org/10.1260/174830108784646661>